MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSCCFS201 -COMPUTER ORGANIZATION

Duration: 1.5 hrs MaximumMarks:50

Students should attempt at least one question from each course outcome to enhance their overall outcome attain ability.

Part A

Short Answer Type Questions

Answer **any 10** Questions

Each question carries **1** mark

1. Define the term “data” in the context of computer systems.[Understand][1]
2. Identify whether MS Paint is system software or application software. [Understand][1]
3. Name the memory that loses content when power is off. [Understand][1]
4. Name the bus used to carry instructions or data to/from the processor. [Understand][1]
5. Name the unit within the CPU that performs logical operations.Understand][2]
6. Write the memory part that stores return addresses during function calls? [Apply][2]
7. Identify the instruction type : JMP 200[Apply][2]
8. Inspect the type of instruction responsible for branching. [analyse][2]
9. State the type of ROM that can be erased and reprogrammed using UV light. [Understand][3]
10. Analyze how increasing the number of address lines affects memory capacity.[analyse][3]
11. Differentiate between SRAM and DRAM based on speed and application.[Analyse][3]
12. Differentiate MDR’s behavior during memory read and memory write.[Analyse [3]

[10x1=10]

Part B

Short Essay Type Questions

Answer **any 4** Questions

Each question carries **5** marks

13. List and describe the major functional units of a computer. [Understand][1]

14. Describe about the bus structure of a computer system

[Understand][1]

15. Discuss the different types of instructions used in computer organization and explain their purposes.[Understand][2]

16. Analyze the impact of PUSH and POP operations on the stack and stack pointer during subroutine execution. Compare how improper use of these operations can affect program flow [Analyse] [2]

17. Compare how different levels of memory in the hierarchy affect the overall performance of a computer system. Provide examples where improper use can slow down the system. [Analyse][3]

18. Illustrate steps for register transfer concepts explaining how a value is moved from register R2 to R4.[Analyse][3]

[4x5=20]

Part C

Essay Type Questions

Answer **any 2** Questions

Each question carries **10** marks

19. Describe in detail, about instruction execution and straight line sequencing. [Understand][1]

20. Compare different addressing modes used in instruction execution with proper examples.

[Analyse][2]

21. Differentiate between RAM and ROM based on their functionality, usage, types, and role in computer systems. Analyze how each type contributes to system performance. [Analyse][3]

[2x10=20]